# DHIKSHA NANDHINI S

**Software Engineer – Data Engineering**

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## PROFESSIONAL SUMMARY

Data-focused Software Engineer with 3+ years of combined experience building real-time and batch data pipelines that process 10M+ telemetry events per day. Hands-on with Python, SQL, Apache Kafka, Apache Flink, Databricks/PySpark, Apache Airflow, dbt, Snowflake, PostgreSQL, and FastAPI across Azure and AWS. Builds event-driven ingestion with event-time handling, late-event processing, idempotent writes, and replay-safe recovery, plus Bronze/Silver/Gold layers and star-schema warehouses for analytics and reporting.

## SKILLS

**Programming:** Python, SQL

**Streaming & Messaging:** Apache Kafka, Apache Flink, Azure Event Hubs, Event-Time Processing

**Data Processing & Orchestration:** Databricks, PySpark, Apache Airflow, dbt, ETL/ELT, Batch & Incremental Processing

**Databases & Warehousing:** PostgreSQL, MongoDB, Snowflake, Apache Doris, Azure Cosmos DB

**Cloud & DevOps:** Microsoft Azure, AWS (S3), Docker, Jenkins, Git, GitHub, CI/CD

**Data Architecture & Modelling:** Medallion Architecture, Star Schema, Fact & Dimension Models, Data Modeling, Data Quality, Schema Validation, Deduplication

**Backend & APIs:** FastAPI, REST APIs, JWT Authentication

**Analytics & Visualization:** Power BI

## PROFESSIONAL EXPERIENCE

**Software Engineer | Vantiva | Bangalore, India** **Jun 2024 – Present**

- Designed and delivered real-time and batch data pipelines using Python, SQL, Apache Kafka, Apache Flink, Databricks/PySpark, and Azure Event Hubs, handling **10M+ telemetry events per day**.
- Orchestrated **ETL/ELT workflows with Apache Airflow and dbt** into Snowflake, improving **data load performance by ~40%** through benchmarking and pipeline tuning.
- Engineered **fault-tolerant ingestion** with event-time handling, idempotent writes, deduplication, and replay-safe recovery, plus **Bronze/Silver/Gold layers** and **star-schema models** for analytics.
- Monitored **Kafka consumer lag and Flink job health**, troubleshooting ingestion delays and adding health checks, metrics, structured logs, and alerts.
- Built secure **FastAPI REST services** (Azure API Management) and partnered with product, mobile, and analytics teams to deliver reliable, analytics-ready datasets.

**SDE Intern | Vantiva | Bangalore, India** **Jan 2024 – May 2024**

- Contributed to device-event correlation workflows for home-network platforms, processing cloud-connected device telemetry into a PostgreSQL-backed device inventory.
- Assisted in building automated test cases and CI integrations for nightly soak and sanity runs.

**SDE Intern | CommScope | Bangalore, India** **May 2023 – Dec 2023**

- Performed **sanity and functional testing** on software builds, validating features against requirements and reporting defects with clear reproduction steps.
- Wrote **SQL queries** to validate test data and backend records during release verification.
- Collaborated with **Android development teams** to reproduce, triage, and verify fixes for reported issues.

## KEY PROJECT CONTRIBUTIONS

**Automated Monthly Broadband Usage Statements**

- Designed an end-to-end **ETL data processing pipeline** supporting an ecosystem of **connected home gateways**, with **1M+ device usage events per day**.
- Implemented **event-driven data ingestion** from Azure Event Hubs, incorporating event deserialization, schema normalization, data validation and structured PostgreSQL loading.

- Developed a **relational data model** separating customer master data, granular device-level usage facts and pre-aggregated monthly consumption summaries.
- Implemented **idempotent data ingestion, deduplication and conflict handling** using PostgreSQL unique constraints to ensure consistency during event reprocessing.
- Developed **SQL-based aggregation and time-window processing** for daily, weekly and monthly consumption analytics, including month-over-month trend analysis.
- Implemented **calendar-based batch processing and incremental data transformations** to generate customer-level broadband usage summaries for automated monthly statements.

**Screen Time Analytics Platform — Android TV**

- Designed an end-to-end **real-time data engineering pipeline** for Android TV's Screen Time feature, processing **10M+ telemetry events per day** from the operator's telemetry service using **Azure Event Hubs → Databricks/PySpark → Cosmos DB → FastAPI**.
- Engineered fault-tolerant stream processing with event-time handling, deduplication, late-event processing, and replay-safe recovery to ensure accurate screen-time calculations without double-counting.
- Developed secure, low-latency **REST APIs** with JWT authentication, per-device authorization, and Azure API Management to deliver application usage insights.

**Subscription Data Warehouse and Analytics Platform**

- Engineered an **AWS S3-to-Snowflake pipeline** orchestrated with Apache Airflow for subscription data.
- Developed **dbt transformations** and star-schema models for analytical reporting.
- Implemented automated **data quality checks** and error handling for reliable data processing.
- Improved data **load performance by 40%** through benchmarking and pipeline optimization.

**STB Telemetry Streaming & Analytics Platform**

- Maintained **Kafka–Flink streaming pipelines** for set-top-box telemetry, supporting event validation, deduplication, and event-time processing.
- Monitored Kafka consumer lag and Flink job health, troubleshooting ingestion delays and processing failures.
- Supported downstream data flow from **Flink → MinIO → Apache Doris**, monitoring data ingestion and reconciliation.
- Maintained **FastAPI and Airflow services** supporting Doris-backed dashboards and data pipeline operations.

**Payment Reconciliation Data Platform**

- Built an event-driven **Kafka pipeline** to ingest and reconcile payment events across systems.
- Implemented **Bronze/Silver/Gold** transformations with schema validation, deduplication and data-quality checks.
- Engineered reliable ingestion using **at-least-once delivery, idempotent writes, retries and quarantine handling**.
- Automated batch reconciliation and reporting, reducing manual effort by 85%.

## CROSS-FUNCTIONAL COLLABORATION

- **Safe Aura:** Android TV parental-control app. Idea Blitz 2025 winner.
- **AI Subtitle Translation:** Real-time subtitle translation POC using on-device ML.

## EDUCATION

**Bachelor of Engineering, Computer Science & Engineering** **2020 – 2024**
Kongu Engineering College, Erode | CGPA: 9.3

## CERTIFICATIONS & AWARDS

- Vantiva Q3 Triumph Award (2026)
- Idea Blitz 2025 — “Safe Aura” (now a deployed client project)
- AWS Cloud Solutions Architect (Coursera), Google Data Analytics Certificate (Coursera)